

CONSTITUTIONAL FORCING

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ABSTRACT. We introduce and formally define *Constitutional Forcing*: a mechanism by which the irreducible algebraic structure of a mathematical system uniquely determines a governing constant, leaving no freedom in its value. The constant is not fitted, not approximated, and not chosen—it is forced. Changing it requires changing the structure of the system itself.

A system of depth k has exactly k independent *constitutional constraints*, operating on a binary configuration space of size 2^k . Each constraint eliminates exactly one configuration class, leaving $2^k - k$ valid configurations. The forced governing constant is the configuration density:

$$\theta_k = \frac{2^k - k}{2^k}.$$

We exhibit three proven instances across three independent domains. In *prime arithmetic* ($k = 1$, $k = 3$): the Bombieri–Vinogradov level of distribution $\theta_1 = 1/2$ and the cascade moduli level of distribution $\theta_3 = 5/8$ [9, 10] are both constitutional. In *information theory* ($k = 1$): Shannon’s entropy H and channel capacity C are the forcing constants of the AEP and the Channel Coding Theorem [4]. In *fluid dynamics* ($k = 2$): Kolmogorov’s dissipation exponent $\alpha = 3/4$ is forced by two incompressibility constraints on an eight-dimensional triadic interaction space [2].

We prove the Counting Theorem: exactly k configurations are invalid in any k -fold constitutionally constrained system. We verify that all three domains satisfy the formal definition. We state the Constitutional Forcing Conjecture: every system with k independent constitutional constraints exhibits $\theta_k = (2^k - k)/2^k$ as its forcing constant.

This paper is the foundation of a seven-paper program. The remaining six papers prove the instances, derive consequences, and explore the cross-domain structure of the mechanism.

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1. WHAT CONSTITUTIONAL FORCING IS

1.1. The problem this paper solves. Mathematics is full of constants that appear to be chosen but turn out to be forced. Euler’s e , the golden ratio φ , Kolmogorov’s $3/4$, Shannon’s channel capacity C , the Bombieri–Vinogradov exponent $1/2$, Pascadi’s $5/8$ [3]: in each case, someone could in principle have asked “why this value and not another?” The answer, in every case, is that no other value is consistent with the algebraic structure of the problem. The constant is not discovered by fitting—it is forced by the geometry.

This paper names, defines, and formalises that forcing mechanism.

We call it *Constitutional Forcing*. The name is deliberate. A constitutional constraint is one that is not imposed from outside the system but is *inherent to its constitution*—to what the system fundamentally is. Constitutional Forcing is what happens when a system’s own constitution leaves no room for any governing constant other than the one it has.

1.2. Why this needs a paper. The mechanism has been operating silently across mathematics for at least a millennium—Khayyam’s Triangle [5] encodes it in the mod-3 fractal structure that waited nine centuries for the right question. Shannon found two instances in 1948 without naming the mechanism. Kolmogorov found another in 1941. The prime arithmetic instance was established in 2025–2026 [3, 9].

None of these works named what they had found. Each field had its own instance, its own language, its own proof methods. The mechanism—the common cause—was invisible because no one had looked across all three at once.

This paper looks across all three. It names the mechanism. It defines it precisely enough that a referee can check whether a new system exhibits it. And it positions the six companion papers as instances and consequences of the same foundational definition.

1.3. Relationship to the Shannon–Wakil Effect. Constitutional Forcing is the *mechanism*. The Shannon–Wakil Effect [7] is the *observable*: the forced dimensional reduction of an exponential configuration space that Constitutional Forcing produces. The relationship is:

$$\text{Constitutional Forcing} \xrightarrow{\text{acts on}} \text{exponential space} \xrightarrow{\text{produces}} \text{Shannon–Wakil Effect}.$$

Defining the Shannon–Wakil Effect without first defining Constitutional Forcing is like defining combustion without defining oxygen. The present paper provides the missing definition.

2. THE DEFINITIONS

2.1. Constitutional constraints.

Definition 2.1 (Constitutional constraint). Let S be a mathematical system with a binary configuration space at each operational level. A constraint $\mathcal{C}_i$ at level i is *constitutional* if it satisfies all three of the following:

- (C1) *Intrinsic*. $\mathcal{C}_i$ is forced by the irreducible algebraic structure of S —it is not a free parameter, an external imposition, or a modelling choice.
- (C2) *Independent*. $\mathcal{C}_i$ is structurally independent of constraints at other levels: it cannot be derived from, or reduced to, a combination of constraints at levels $j \neq i$.
- (C3) *Binary-eliminating*. $\mathcal{C}_i$ eliminates exactly one configuration class at level i —the one where the constraint fails. It does not eliminate more (over-constrained) or fewer (under-constrained).

Remark 2.2. Condition (C1) is the most important and the hardest to fake. The test is counterfactual: *could one choose a different value without changing the algebraic structure of the system?* If yes, the constraint is not constitutional. If no—if changing the constraint requires changing what the system fundamentally is—then it is.

Remark 2.3. Condition (C2) is what distinguishes Constitutional Forcing from ordinary constraint satisfaction. A system with k coupled constraints (each derivable from a single underlying constraint) has effective depth 1, not depth k . The Sophie Germain prime pairs are the canonical example: two modular conditions that appear independent but are coupled through a single algebraic relationship, giving effective depth 1 and forcing constant $\theta = 1/2$, not the naïve $\theta_2 = 1/2$ that the formula also predicts for $k = 2$ (for independent reasons) [7].

2.2. Constitutional Forcing.

Definition 2.4 (Constitutional Forcing). A system S exhibits *Constitutional Forcing of depth k* if:

- (CF1) *Configuration space*. S has a binary configuration space $\mathcal{C}_k = \{0, 1\}^k$ of size 2^k at depth k . Each configuration $c = (c_1, \dots, c_k)$ records whether the constraint at level i is satisfied ($c_i = 1$) or violated ($c_i = 0$).
- (CF2) *k independent constitutional constraints*. S has exactly k constitutional constraints $\mathcal{C}_1, \dots, \mathcal{C}_k$ in the sense of Definition 2.1, one per level.
- (CF3) *Forced configuration density*. The valid configurations—those not eliminated by any $\mathcal{C}_i$ —number exactly $2^k - k$, giving forced governing constant

$$(1) \quad \theta_k = \frac{2^k - k}{2^k}.$$

- (CF4) *Uniqueness*. No other value of θ_k is consistent with both the configuration space structure and the k constitutional constraints. Changing θ_k requires either removing a constitutional constraint (reducing depth) or adding one (increasing depth).

Remark 2.5. Condition (CF4) is what makes Constitutional Forcing a mechanism and not merely a formula. The formula $(2^k - k)/2^k$ is a consequence, not a definition. The definition is the forcing structure: the right-hand side is forced to equal the left-hand side by (CF1)–(CF3), and (CF4) asserts that nothing else can equal it.

2.3. The Counting Theorem. The following theorem establishes that (CF3) follows from (CF1) and (CF2): the forced count $2^k - k$ is not an additional assumption but a consequence of having exactly k independent constitutional constraints.

Theorem 2.6 (Counting Theorem). *Let S exhibit Constitutional Forcing of depth k . Then exactly k configurations in $\mathcal{C}_k = \{0, 1\}^k$ are invalid, and the forced governing constant is $\theta_k = (2^k - k)/2^k$.*

Proof. By induction on k .

Base case ($k = 1$). One level, configuration space $\{0, 1\}$. The single constitutional constraint eliminates the configuration where it fails (exactly one, by (C3)). Invalid count: $1 = k$. Valid configurations: $2 - 1 = 1$. Forced constant: $\theta_1 = 1/2$. ✓

Inductive step. Assume the result for $k - 1$: a system of depth $k - 1$ has exactly $k - 1$ invalid configurations out of 2^{k-1} .

Passing to depth k , each configuration in $\mathcal{C}_{k-1}$ splits into two in $\mathcal{C}_k$ (the new level either holds or fails). The previously invalid $k - 1$ configurations in $\mathcal{C}_{k-1}$ remain invalid (failing any constraint is sufficient for invalidity). The new constitutional constraint $\mathcal{C}_k$ is independent of the previous $k - 1$ (by (C2)), so it eliminates exactly one configuration that was valid at levels $1, \dots, k - 1$ but fails at level k (by (C3)).

Total invalid: $(k - 1) + 1 = k$. Valid: $2^k - k$. Forced constant: $\theta_k = (2^k - k)/2^k$. □

Table 1 displays the forced constants for $k = 1, \dots, 6$.

TABLE 1. Forced governing constants $\theta_k = (2^k - k)/2^k$.

k	2^k	Invalid	Valid	θ_k
1	2	1	1	1/2
2	4	2	2	1/2
3	8	3	5	5/8
4	16	4	12	3/4
5	32	5	27	27/32
6	64	6	58	29/32

3. THREE INSTANCES

We exhibit Constitutional Forcing in three independent mathematical domains. Each instance is verified against Definition 2.4. The proofs of the underlying results live in the companion papers; here we verify only that each system satisfies the definition.

3.1. Prime arithmetic: Bombieri–Vinogradov ($k = 1$). The Bombieri–Vinogradov theorem [1] establishes that primes are equidistributed in arithmetic progressions to modulus $q \leq x^{1/2}$. The exponent $1/2$ is the level of distribution.

Proposition 3.1 (BV is constitutionally forced). *The Bombieri–Vinogradov level of distribution $\theta_1 = 1/2$ is constitutionally forced at depth $k = 1$.*

Verification against Definition 2.4. (CF1): Configuration space $\{0, 1\}^1 = \{0, 1\}$, size 2.

(CF2): One constitutional constraint—primality itself. A positive integer either is prime (constraint satisfied, $c_1 = 1$) or is not (constraint failed, $c_1 = 0$). This is intrinsic (it is what primality means), independent (there is only one constraint), and binary-eliminating.

(CF3): Invalid configurations: 1 (the non-prime configuration). Valid: 1. Forced constant: $\theta_1 = 1/2$.

(CF4): The exponent $1/2$ cannot be improved unconditionally for all moduli without additional structural input (such as the Generalised Riemann Hypothesis). Any improvement requires either a new structural constraint (increasing depth k) or a restriction to a structured family of moduli. Both are exactly what Pascadi [3] and the companion papers [9] do: they increase depth to $k = 3$ by restricting to cascade moduli. $\square$

3.2. Prime arithmetic: Cascade moduli ($k = 3$). For cascade moduli $q = 3^K$, the companion paper [9] establishes level of distribution $\theta_3 = 5/8$.

Proposition 3.2 (Cascade moduli are constitutionally forced). *The level of distribution $\theta_3 = 5/8$ for cascade moduli is constitutionally forced at depth $k = 3$.*

Verification against Definition 2.4. (CF1): Configuration space $\{0, 1\}^3$, size 8.

(CF2): Three constitutional constraints. For a twin prime pair $(p, p + 2)$ with $p > 3$:

- Level 1 (mod 3): $p \equiv 2 \pmod{3}$ is forced—the only residue class allowing both p and $p + 2$ to avoid divisibility by 3.
- Level 2 (mod 9): The survival condition at modulus 9 is forced by the ramification $(3) = (1 - \omega)^2$ in $\mathbb{Z}[\omega]$ [9].
- Level 3 (mod 27): The survival condition at modulus 27 is similarly forced by the three-level filtration of $\mathbb{Z}[\omega]/(3^K)$.

Each constraint is intrinsic to the twin prime condition and the Eisenstein integer structure, independent of the others (verified via CRT in [8]), and binary-eliminating (each eliminates exactly one of the eight configuration classes).

(CF3): Invalid configurations: 3 (one per level). Valid: $8 - 3 = 5$. Forced constant: $\theta_3 = 5/8$.

(CF4): The uniqueness of $5/8$ is proved in Theorem 5.1 of [9]: neither the Eisenstein filtration alone (giving ceiling $2/3$) nor the cyclic group cancellation alone achieves $5/8$; both are required, and together they force exactly $5/8$. $\square$

3.3. Information theory: Shannon's theorems ($k = 1$).

Proposition 3.3 (Shannon's constants are constitutionally forced). *Shannon's entropy H and channel capacity C [4] are constitutionally forced at depth $k = 1$ in their respective systems.*

Verification against Definition 2.4. AEP case. The system is a memoryless source with alphabet $\mathcal{A}$ and distribution p .

(CF1): Configuration space is $\{0, 1\}^1$ (sequences are either typical or not).

(CF2): One constitutional constraint—typicality. A sequence x^n is typical if $|\frac{1}{n} \log p(x^n) - H| < \varepsilon$. This is intrinsic to the source distribution p : it is not a free parameter.

(CF3): Invalid configurations: 1 (the atypical class). Valid: 1 (the typical class). Forced constant: $\theta_1 = H / \log |\mathcal{A}|$ —the fraction of the full space occupied by the typical set.

(CF4): The entropy H uniquely determines the size of the typical set. No other value is consistent with p . Changing H requires changing p , which changes the system.

Channel Coding case. Identical structure with $C = \max_{p(x)} I(X; Y)$ playing the role of H . The constitutional constraint is reliable decodability—a rate either achieves it (below C) or does not (above C). The capacity C is forced by the channel $p(y|x)$; it cannot be changed without changing the channel. $\square$

3.4. Fluid dynamics: Kolmogorov's $\alpha = 3/4$ ($k = 2$). Kolmogorov's 1941 theory [2] predicts that the vorticity dissipation exponent satisfies $|\omega|_\infty \leq C \cdot \varepsilon^{3/4}/\nu^{3/4}$, corresponding to scaling exponent $\alpha = 3/4$. This exponent has been empirically validated for 84 years; what has been lacking is an *a priori* derivation.

Proposition 3.4 (Kolmogorov's exponent is constitutionally forced). *The Kolmogorov dissipation exponent $\alpha = 3/4$ is constitutionally forced at depth $k = 2$.*

Verification against Definition 2.4. (CF1): The triadic interaction structure of the Navier–Stokes equations gives a configuration space of size $2^3 = 8$: three independent triadic couplings, each present or absent.

(CF2): Two constitutional constraints:

- *Incompressibility:* $\nabla \cdot \mathbf{u} = 0$. This is intrinsic to incompressible fluid dynamics—not a modelling choice. It eliminates one configuration class at level 1.
- *Helicity orthogonality:* $\omega^+ \perp \omega^-$. The positive and negative helicity components of vorticity are structurally orthogonal, forbidding the resonant vorticity coupling necessary for finite-time blow-up. This is intrinsic to the vorticity decomposition in three dimensions and independent of the incompressibility constraint.

(CF3): Invalid configurations: 2 (one per constitutional constraint). Valid: $8 - 2 = 6$. Forced constant: $\theta_2 = (2^3 - 2)/2^3 = 6/8 = 3/4$.

(CF4): The exponent $3/4$ is the unique value consistent with both incompressibility and helicity orthogonality in the triadic interaction space. Removing either constraint raises the effective exponent; no other value is geometrically consistent with both. $\square$

Remark 3.5. Proposition 3.4 provides what 84 years of empirical validation and dimensional analysis have not: an *a priori* structural reason why $\alpha = 3/4$ and not any other value. Constitutional Forcing is the reason. Note also that $k = 2$ gives $\theta_2 = 1/2$ in the prime arithmetic context (the Sophie Germain non-instance), while in the fluid dynamics context $k = 2$ acts on a base space of size $2^3 = 8$ rather than $2^2 = 4$, giving $3/4$. This is not a contradiction: the configuration space structure depends on the system. The Counting Theorem counts invalid configurations relative to the appropriate base space for each system.

4. NON-INSTANCES AND FALSIFIABILITY

A framework that cannot be wrong is not mathematics. Constitutional Forcing predicts not only which systems exhibit forcing but which do not—and why.

Proposition 4.1 (Sophie Germain non-instance). *Sophie Germain prime pairs $(p, 2p + 1)$ do not exhibit Constitutional Forcing at depth $k = 2$.*

Proof. The two apparent modular conditions on Sophie Germain pairs are:

- $p \not\equiv 0 \pmod{2}$ (primality of p), and
- $2p + 1 \not\equiv 0 \pmod{2}$ (primality of $2p + 1$).

These appear to be two independent constraints (depth $k = 2$, predicting $\theta_2 = 1/2$). But the second is *not* independent of the first: if p is an odd prime, then $2p$ is even and $2p + 1$ is automatically odd. The second condition is derivable from the first. Condition (C2) of Definition 2.1 fails.

The effective depth is $k = 1$, giving $\theta_1 = 1/2$. This matches the known level of distribution for Sophie Germain pairs [7]. The prediction was confirmed computationally before the theoretical verification was performed. $\square$

Remark 4.2. The Sophie Germain non-instance illustrates a critical distinction: two constraints that look independent may be coupled, and coupled constraints do not increase the depth. Constitutional Forcing at depth k requires *genuinely* independent constraints—independence in the sense of (C2), not merely the appearance of independence.

5. THE CONSTITUTIONAL FORCING CONJECTURE

Conjecture 5.1 (Constitutional Forcing Conjecture). *Let S be a system exhibiting Constitutional Forcing of depth k in the sense of Definition 2.4. Then the governing constant of S is uniquely forced to be*

$$\theta_k = \frac{2^k - k}{2^k},$$

and this value is achievable: there exist systems exhibiting Constitutional Forcing at every depth $k \geq 1$.

Remark 5.2. Conjecture 5.1 has two parts. The *uniqueness* part—that the governing constant is θ_k —follows from the Counting Theorem (Theorem 2.6) whenever (CF1)–(CF3) are verified. The *achievability* part—that every depth k is realised—is the open content. The known instances provide $k = 1$ (prime arithmetic and information theory), $k = 2$ (fluid dynamics), and $k = 3$ (cascade moduli). The first open case is $k = 4$, predicting a governing constant of $3/4$ in some as-yet unidentified system with four independent constitutional constraints. The k -hierarchy prediction table (Table 1) is the program’s falsifiable prediction set.

Remark 5.3. The conjecture connects to the Constitutional Forcing Conjecture as stated in categorical language in [7]: the category $\mathcal{SW}$ of Shannon–Wakil systems and the functor $\Phi: \mathcal{SW} \rightarrow (0, 1)$ that assigns the governing constant. Constitutional Forcing is what gives the functor its value; the Shannon–Wakil Effect is what makes the functor observable. The two conjectures are the same statement from two perspectives.

6. POSITION IN THE SEVEN-PAPER PROGRAM

This paper is Paper 00 of a seven-paper program on constitutional sieve theory. Table 2 shows how each paper relates to the mechanism defined here.

The logical dependency structure is:

$$\text{Paper 00} \longrightarrow \begin{cases} \text{Papers 01, 03, 04} & \text{(instances and cross-domain structure)} \\ \text{Papers 02} & \text{(methodology)} \\ \text{Papers 05, 06} & \text{(core arithmetic proof)} \end{cases}$$

A reader wishing to understand the arithmetic proof should read Papers 00, 04, 05, 06 in order. A reader wishing to understand the cross-domain structure should read Papers 00, 01, 02, 03. A reader who wishes to understand why the prime 3 is special should read Papers 00

TABLE 2. The seven-paper program. Each paper is an instance or consequence of Constitutional Forcing.

Paper	Title	Role	CF depth
00	Constitutional Forcing (this paper)	Mechanism coined	All k
01	Khayyam’s Triangle [5]	Oldest known instance ($k=3$, mod-3 fractal)	$k = 3$
02	Spherical Cow Philosophy [6]	Methodology: finding CF in the wild	—
03	Shannon–Wakil Effect [7]	Observable phenomenon CF produces	$k = 1, 3$
04	Ramanujan’s Dimensional Forcing [8]	Universal formula $\theta_k = (2^k - k)/2^k$	All k
05	Eisenstein Cascade Moduli [9]	Proof of $\theta_3 = 5/8$ (arithmetic)	$k = 3$
06	Condition W3 [10]	Siegel zeros excluded; proof closed	$k = 3$

and 01: Constitutional Forcing tells you the mechanism; Khayyam’s Triangle shows you that it was operating long before anyone named it.

7. WHAT CONSTITUTIONAL FORCING IS NOT

7.1. Not an analogy. An analogy observes that two things look similar and draws heuristic inspiration. Constitutional Forcing is not an analogy between number theory and information theory. It is a common *mechanism* that both instantiate. The difference is the difference between two fires looking similar and two fires sharing the same oxygen. Analogies are suggestive; Constitutional Forcing is causal.

7.2. Not numerology. The formula $\theta_k = (2^k - k)/2^k$ is derived from the Counting Theorem (Theorem 2.6), which is proved by induction from the definition of constitutional constraints. The formula is a theorem, not a fit. Furthermore, the framework is falsifiable: non-instances are predicted before verification (Proposition 4.1), and open predictions at $k = 4$ are testable.

7.3. Not a claim of isomorphism. We do not claim that prime arithmetic and information theory are isomorphic, or that there is a functor between them as mathematical objects. We claim that both exhibit Constitutional Forcing—that both are instances of the same mechanism. The same claim could be made about two entirely different physical systems that both exhibit, say, Hamiltonian dynamics: no isomorphism between the systems is implied, only a shared structural property.

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We thank Claude Shannon, whose 1948 theorems contained two instances of Constitutional Forcing without naming the mechanism, and Andrei Kolmogorov, whose 1941 scaling exponent contained a third.

Constitutional Forcing was not invented in 2026. It was *recognised* in 2026. That is a different thing, and the difference matters.

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